

Bluestock Mutual Fund Analytics Capstone Project

Executive Summary

The Indian mutual fund industry has experienced significant growth in recent years, driven by increasing investor participation, rising SIP contributions, and growing awareness of financial planning. The objective of this project was to build an end-to-end analytics solution capable of transforming raw mutual fund datasets into actionable business insights.

The project covered the complete analytics lifecycle, including data engineering, exploratory data analysis, performance measurement, risk modeling, advanced analytics, and dashboard development. More than 40 mutual fund schemes and multiple industry datasets were analyzed using Python, SQLite, and Power BI.

The final solution enables stakeholders to evaluate fund performance, understand investor behavior, monitor industry trends, and identify risk-adjusted investment opportunities.

1. Introduction

Mutual funds have become one of the most preferred investment vehicles in India due to professional fund management, diversification benefits, and accessibility through SIP investments.

However, analyzing large volumes of mutual fund data manually is challenging. Fund houses, analysts, and investors require data-driven tools to evaluate performance, risk, and investor participation trends.

This project aims to address these challenges by developing a comprehensive mutual fund analytics platform.

2. Business Problem

The mutual fund industry generates large volumes of data across NAV histories, investor transactions, portfolio holdings, benchmark indices, and AUM statistics.

Without centralized analytics, stakeholders face difficulties in:

- Tracking fund performance
- Measuring risk-adjusted returns
- Understanding investor behavior
- Comparing schemes against benchmarks
- Identifying emerging investment trends

The project was designed to provide a unified analytical framework to solve these challenges.

3. Project Objectives

1. Build an automated ETL pipeline.
 2. Create a structured analytical database.
 3. Perform exploratory data analysis.
 4. Calculate advanced fund performance metrics.
 5. Develop interactive business dashboards.
 6. Build risk analytics models.
 7. Generate investment insights and recommendations.
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4. Data Sources

The project utilized multiple datasets:

- Fund Master Dataset
- NAV History Dataset
- AUM Dataset
- SIP Inflow Dataset
- Category Inflow Dataset
- Investor Transactions Dataset
- Portfolio Holdings Dataset
- Benchmark Indices Dataset

Together, these datasets provided a complete view of industry activity and fund performance.

5. ETL Architecture

The ETL pipeline was implemented using Python and Pandas.

Key activities included:

- Data ingestion
- Missing value treatment
- Data validation
- Standardization
- Duplicate removal
- Export to processed datasets

SQLite was used to store cleaned and structured data for downstream analytics.

(Insert ETL Architecture Diagram)

6. Data Cleaning Methodology

Several data quality issues were identified and corrected:

- Missing NAV values
- Duplicate records
- Inconsistent transaction labels
- Invalid transaction amounts
- Incorrect date formats

Data quality validation rules ensured consistency across all datasets.

7. Exploratory Data Analysis

EDA revealed several industry trends.

NAV Analysis

Most schemes demonstrated consistent NAV growth during the observation period, reflecting overall market expansion.

(Insert NAV Trend Chart)

SIP Analysis

SIP inflows showed strong growth throughout the period, indicating increasing retail investor participation.

(Insert SIP Trend Chart)

Category Analysis

Flexi Cap, Mid Cap, and Small Cap categories attracted significant investor inflows.

(Insert Category Heatmap)

Demographic Analysis

Younger investors demonstrated increasing participation in mutual fund investments.

(Insert Age Distribution Chart)

8. Performance Analytics

Performance evaluation was conducted using several quantitative metrics.

CAGR

Annualized growth rates were calculated across all schemes.

Sharpe Ratio

Risk-adjusted returns were measured using the Sharpe Ratio framework.

Sortino Ratio

Downside risk-adjusted performance was evaluated through Sortino Ratios.

Alpha and Beta

Benchmark-relative performance was analyzed using regression-based Alpha and Beta metrics.

Maximum Drawdown

Worst historical declines were identified for each fund.
(Insert Fund Scorecard Table)

9. Dashboard Development

An interactive Power BI dashboard was developed.

The dashboard consists of four pages:

1. Industry Overview
2. Fund Performance
3. Investor Analytics
4. SIP & Market Trends

(Insert Dashboard Screenshots)

The dashboard allows users to interactively explore trends and compare schemes across multiple dimensions.

10. Advanced Analytics

Historical VaR and CVaR

Downside risk was measured using Value at Risk and Conditional Value at Risk models.

Rolling Sharpe Ratio

Risk-adjusted performance trends were evaluated over rolling 90-day windows.

Investor Cohort Analysis

Investor groups were segmented based on first transaction year.

SIP Continuity Analysis

Investor retention and SIP discipline were measured through transaction frequency analysis.

HHI Concentration Analysis

Portfolio diversification was evaluated using Herfindahl-Hirschman Index calculations.

Fund Recommendation Engine

A rule-based recommendation model was developed to suggest funds based on risk preferences.

11. Key Findings

1. SIP inflows displayed strong and consistent growth.
 2. Equity-oriented schemes dominated investor preference.
 3. Risk-adjusted performance varied significantly across funds.
 4. Investor retention remained strong across SIP participants.
 5. Portfolio concentration differed considerably among schemes.
 6. Some funds generated positive alpha while maintaining controlled downside risk.
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12. Recommendations

- Promote SIP adoption among younger investors.
 - Monitor highly concentrated portfolios.
 - Incorporate risk-adjusted metrics into investment decisions.
 - Expand benchmarking beyond Nifty 50 and Nifty 100.
 - Enhance investor education programs.
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13. Limitations

- Synthetic educational dataset.
 - Limited historical period.
 - Simplified recommendation logic.
 - Absence of real-time NAV updates.
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14. Future Enhancements

- Streamlit Web Application
 - Automated NAV ingestion via APIs
 - Monte Carlo Portfolio Simulation
 - Markowitz Portfolio Optimization
 - Automated Weekly Reporting
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15. Conclusion

The project successfully delivered a complete mutual fund analytics ecosystem combining data engineering, financial analytics, business intelligence, and risk management. The resulting solution demonstrates how data-driven approaches can improve investment decision-making and provide valuable insights into mutual fund industry trends.
